

## Use and maintenance manual for generator

### MR\VP 10K-CH

N° 70456-70458\C.3411\07

GENFRAI VACUUM EQUIPMENT Ltd



#### **Warning!**

*This operations manual required for the safe use of the generator. It is therefore advisable to keep it near the unit.*

| 1    | 09/01/08 | First edition |             | Mario Della Santa |          |
|------|----------|---------------|-------------|-------------------|----------|
| Rev. | Date     | Description   | Compilation | Check             | Approval |

This manual is directed to persons working with the MR\VP generator, in particular to personnel involved in its operation.

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This documentation contains a description of the product that is as accurate as possible.

The product may be subject to technical changes without previous notice.

For any questions or problems regarding installation, contact Me.Ro. srl. We will be pleased to assist you.

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## 1 GENERAL INFORMATION

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*Who is this chapter  
directed at?*

### Profile of this chapter:

This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.

*Contents of the chapter*

This chapter contains the main details of the product and of the manual.

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### 1.1 PRODUCER'S WARRANTY

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#### WARRANTY CONDITIONS:

Me.Ro. srl's equipment is built using tested materials. Once manufactured, the equipment is further tested.

*12-month warranty*

Our equipment is guaranteed **12 months** ex-works our factory, located in Ponte a Moriano, Lucca (Italy). In case of requests for intervention on the equipment during the warranty period, travel hours and costs, food and lodging will be invoiced to the requesting company. Working hours spent to repair the problem and any spare parts necessary are to be borne by Me.Ro. srl.

#### EXCLUDED FROM WARRANTY:

Damages to the equipment caused by an improper use or by unauthorized operation thereof are excluded from warranty.

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## 1.2

## STRUCTURE OF THE DOCUMENTATION

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### GENERAL

Two types of documents are supplied together with *Me.Ro. srl*'s products:

- Use and maintenance manual  
This manual contains all details necessary for the unit's installation, operation and maintenance.
- Technical documents concerning the unit  
The technical documentation regarding the unit includes layout, drawings and materials list.

This document is the **use and maintenance manual**.

### 1.3

### SYMBOLS LEGENDA



#### **Danger!**

*This signal warns of possible physical injury.*



#### **Danger: high voltage**

*This signal warns of a danger deriving from the presence of high voltage. If not respected, electrical shock may ensue.*



#### **Danger: High frequency!**

*This signal warns of a danger deriving from high frequency radiation. People wearing pacemakers or prostheses should not access the area bearing this symbol. The pacemaker's operation could be disturbed and prostheses' internal temperatures could rise.*



#### **Warning!**

*This symbol calls attention to important information.*

### 1.4

### TECHNICAL EXPRESSIONS AND ABBREVIATIONS

#### DESCRIPTION OF ABBREVIATIONS

|     |            |
|-----|------------|
| $P$ | Power      |
| $I$ | Current    |
| $V$ | Voltage    |
| $f$ | Frequency  |
| $L$ | Inductance |
| $C$ | Capacity   |

## 2

## SAFETY

### Profile of this chapter:

*Who is this chapter directed at?*

This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.

*Contents of the chapter*

This chapter contains the main details for safe operation of the product.

### 2.1

### INTENDED USE



#### **Danger!**

*Protection of people and of the system is not guaranteed if the generator does not work in accordance with its intended use.*



**Any maintenance and repair operations on the electronic generator and on the electrodes must be performed exclusively by authorized specialized personnel.**

*Plasma applications*

#### INTENDED USE

The generator is conceived as an energy source to supply a vacuum chamber for plasma formation.

The generator is to be used exclusively for its intended use.

### 2.2

### GENERAL WARNINGS

#### DANGER: PRESENCE OF HIGH VOLTAGE



#### **Danger: high voltage!**

*The generator converts network voltage in **DC** and subsequently in the inverter stage in **AC**. Under these voltages, the components can cause electric shock.*



**Before intervening on the generator and on the plasma chamber, turn the general switch to OFF.**

## 2.3

## DANGEROUS AREAS: MAINTENANCE AND REPAIR

DANGEROUS AREAS: MAIN SWITCH TO OFF



**Remove power supply before intervening on the generator.**

## 2.4

## SAFETY DEVICES

### CONTROL AND PROTECTION DEVICE

#### Warning!



*The operator is responsible for safety operations on the generator. He/she should be acquainted with the norms and apply them.*

The electronic generator comes with a general switch and an external emergency button or device.

The external emergency button or device can be connected to the relative terminals located inside the generator (see electrical diagram) by disabling the actuation of the contactors that supply the generator itself.



#### Danger!

*The general switch does not guarantee personal safety and in any case must not be used as an emergency switch.*

*Before intervening on the generator, wait about a 30" to allow the DC link to discharge.*

### Protection systems

The generator is immediately turned OFF in case of:

- Overvoltage
- Overheating
- Overloading
- Interlocks
- General switch open

## 2.5

## SAFETY MEASURES



**In case of emergency, respect the following steps in the given order:**

1. Interrupt the generator's external supply line and any auxiliary supplies.
2. Extinguish the fire through an appropriate extinguisher.



#### **Danger: high voltage!**

*Due to the presence of high voltage, never extinguish a fire with water but use appropriate extinguishing agents instead.*



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## 3 TECHNOLOGICAL DESCRIPTION

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### Profile of this chapter:

*Who is this chapter  
directed at?*

This chapter is directed to persons who install, operate and perform maintenance work on the electronic generator.

*Contents of the chapter*

This chapter contains information on generators' operation and technology, uses and application fields and technical data.

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### 3.1 THE COMPONENTS AND THEIR FUNCTIONS

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#### GENERATOR CONSTRUCTION

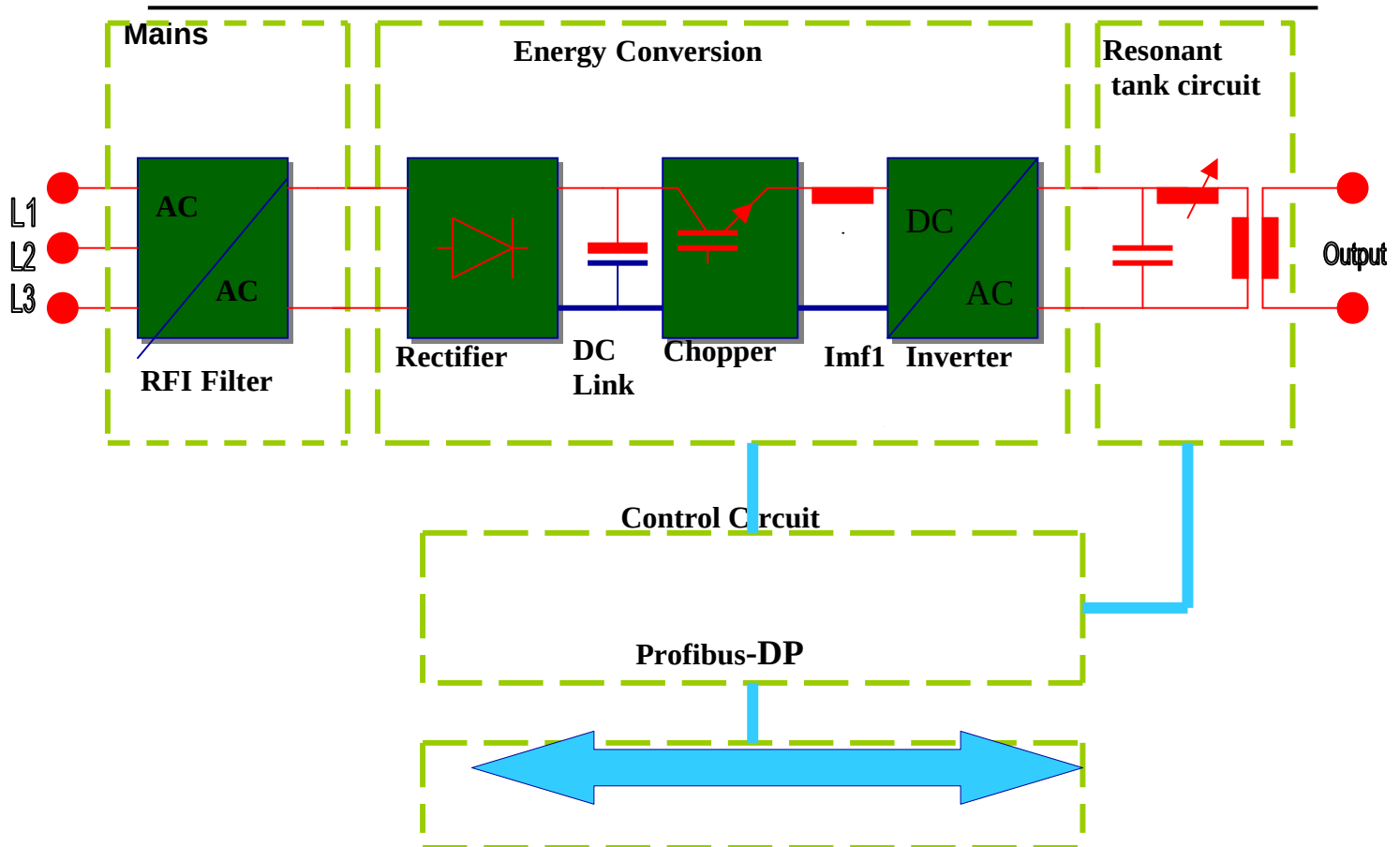
The generator is built in metallic cabinet.

Energy conversion components **IGBT – Diodes – Resonant tank circuit** are air-cooled.

#### COMPONENTS

The figure below illustrates the generator's diagram in blocks. Essentially, it is comprised of:

- **Mains**
- **Rectifier**
- **Chopper**
- **Inverter**
- **Output transformer**
- **Control electronics**
- **Arcs management**
- **Profibus DP**



**Mains:** this stage is comprised of an RFI filter that protects the equipment from network interferences and respects the requirements of the norm on electromagnetic compatibility, “EMC”, for this type of equipment.

**The rectifier** converts alternate network voltage into continuous voltage to supply the Chopper stage.

**The Chopper** regulates the generator’s output power, controls current and voltage limitations and, through impedance IMF1, supplies the inverter.

**The Inverter** is comprised of four IGBT modules in full-bridge configuration, is supplied in current and coupled with a resonant circuit comprised of the output transformer’s primary inductance and by the CVOL capacity.

**The output transformer**, as mentioned above, is an integral part of the inverter; multi-polar secondary winding allows easy adaptation between generator and load.

**The control electronics** perform all the functions related to the control and piloting of the generator’s power stages, as well as the management of the arcs that are created inside the plasma chamber.

**Arcs management:** when an arc is formed inside the plasma chamber, its load resistance drastically decreases, causing an increase in the output current lout.

If the increase exceeds a determinated set value (value that changes proportionally with the Power Level in the used function mode), the generator is turned to OFF and, after a brief delay, is reset automatically to ON.

---

When the number of arcs exceeds the set value (*Number of arcs, VW2018 Profibus-DP*) in the opening time of the arcs count window (*ArcCountWin., VW2020 Profibus-DP*), a message alarm is displayed: "Arcs limit reached" and the generator is set to OFF. See subsequent table.

If the number of arcs in the count window opening time is less than the set value, the message alarm "Arcs limit reached" is not displayed and the counter is reset upon count window closure

**The Profibus-DP** allows to remote-control the generator from a supervision device that manages the system.

## 3.2

## CONTROL MODE

### PROFIBUS-DP. EM277

#### Configuration 6

32 Word Master inlet

32 Word Master outlet

#### PROFIBUS-DP GENERAL VACUUM – ME.RO Interface table

| MASTER    | GEN.VAC. | WRITING TO ME.RO         | PLC S7 200 SIEMENS CPU 224            | SLAVE   | ME.RO |
|-----------|----------|--------------------------|---------------------------------------|---------|-------|
| PAW X     | Bit 0    | GENERATOR.<br>START-STOP | 1 = START 0 = STOP                    | V2000.0 |       |
|           | Bit 1    | RESET                    | 1 = RESET (Impulse)                   | V2000.1 |       |
|           | Bit 1    | RESET                    | 1 = RESET TOTAL ARCS (old down 3 sec) | V2000.1 |       |
|           | Bit 2    | REMOTE CONTROL           | 1 = REMOTE ENABLED                    | V2000.2 |       |
|           | Bit 3    | SEL MANUAL               | 1 = MANUAL MODE SELECTED              | V2000.3 |       |
|           | Bit 4    | SEL AUT                  | 1 = AUT MODE SELECTED                 | V2000.4 |       |
|           | Bit 5    | SEL AUT SPEC POWER       | 1 = AUT SPEC POWER MODE SELECTED      | V2000.5 |       |
| PAW X +2  |          | LOW SPEED                | LOW SPEED (m/min)                     | VW2002  |       |
| PAW X +4  |          | POWER MAX DEVIATION      | MAX. POWER DEVIAT. ADMISSIBLE %       | VW2004  |       |
| PAW X +6  |          | PLASMA CHAMBER<br>LENGTH | PLASMA CHAMBER LENGTH (cm)            | VW2006  |       |
| PAW X +8  |          | SP MAN.                  | 0 – 100 %                             | VW2008  |       |
| PAW X +10 |          | SP AUT                   | 1.000 – 10.000 W.                     | VD2010  |       |
| PAW X +14 |          | SP AUT SPEC POWER        | 5 – 50 (Wmin/m <sup>2</sup> )         | VW2014  |       |
| PAW X +16 |          | SPEED                    | SPEED OF WORK                         | VW2016  |       |
| PAW X +18 |          | NUMBER OF ARCS           | SELECT NUMBER OF ARCS                 | VW2018  |       |
| PAW X +20 |          | TIME FOR ARCS            | TIME FOR ARCS IN SECONDS              | VW2020  |       |
| PAW X +22 |          | TAP USED                 | OUT. TRANSF. TURNS USED               | VW2022  |       |

| MASTER    | GEN.VAC   | READ BY ME.RO  | S7 200                                         | SLAVE   | ME.RO |
|-----------|-----------|----------------|------------------------------------------------|---------|-------|
| PEW X     | Bit 0     | GENERATOR ON   | 1 = GENERATOR ACTIVE                           | V2064.0 |       |
|           | Bit 1     | ALARM 1        | 1 = ALARM GENERATOR                            | V2064.1 |       |
|           | Bit2      | ALARM 2        | 1 = POWER DEVIATION ALARM                      | V2064.2 |       |
|           | Bit 3 -15 |                | Spare                                          |         |       |
| PEW X +2  |           | I. DC          | GENERATOR WORKING CURRENT (A.)                 | VD2066  |       |
| PEW X +6  |           | V.DC           | GENERATOR WORKING VOLTAGE (V.)                 | VD2070  |       |
| PEW X +10 |           | V.OUT          | V. Out rms (Plasma Chamber)                    | VW2074  |       |
| PEW X +12 |           | I.OUT          | A.Out rms (Plasma Chamber)                     | VD2076  |       |
| PEW X +16 |           | P.OUT          | GENERATOR POWER OUTPUT (W.)                    | VD2080  |       |
| PEW X +20 |           | SPEC POWER ACT | GEN. ACTUAL SPEC. POWER (Wmin/m <sup>2</sup> ) | VD2084  |       |
| PEW X +24 |           | ALARM          | WORD ALARM                                     | VW2088  |       |
| PEW X +26 |           | SIGNALLING     | WORD SIGNALLING                                | VW2090  |       |
| PEW X +28 |           | ARCS           | NUMBER OF ARCS                                 | VW2092  |       |
| PEW X +30 |           | SPEED          | LINE SPEED m/min                               | VW2094  |       |

#### **WORD ALARM - VW2088 -**

Word VW2088 assumes different values based on the alarm:

**VW2088 = 0 Ready to start**

**VW2088 = 1 Low Speed**

**VW2088 = 2 Emergency**

**VW2088 = 4 Vcc Fault**

**VW2088 = 8 Thermal overload**

**VW2088 = 16 Cooling water plasma chamber**

**VW2088 = 32 Metallizer open**

**VW2088 = 64 Overload**

**VW2088 = 512 Arc limit reached**

#### **WORD SIGNALLING – VW2090**

Word VW2090 assumes different values according to the signalling to be made:

**VW2090 = 1 Ready to start**

**VW2090 = 4 Generator ON**

**VW2090 = 8 Low speed**

**VW2090 = 16 Present alarm**

**VW2090 = 32 Power too high**

**VW2090 = 64 Power too low**

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## 4 INSTALLATION

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### Profile of this chapter:

This chapter is directed to persons who install the electronic generator.

*Who is this chapter  
directed at?*

*Contents of the chapter*

This chapter contains information concerning the proper mechanical and electronic installation of the generator and the station.

---

### 4.1 TRANSPORT, PLACEMENT

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#### TRANSPORT

Once you receive the equipment:



**Check the integrity of all its parts so as to verify any damage that may have been caused by transport.**



**Immediately signal any damage that may have been caused by transport.**

For shipment, the system's components are packed in appropriate containers having a wooden base.



**We advise that you remove the packaging only when the machine is near to its definitive position.**

The machine may be moved with a forklift of suitable capacity.

The gross weight is reported on the packaging.

To move the machine without its packaging, both the generator and the discharge stations must be lifted using pre-arranged ring-bolts.

#### PLACEMENT

*Environmental  
conditions*

The machine must be adequately protected against water and dust.

It may be warehoused at temperatures included between -25°C and +70°C.

A dry environment is preferable (not mandatory).

---

### 4.2 UNPACKING

---



**Carefully unpack the various components of the generator**

The packing contains all the cables necessary to start the machine's operation, except for the main supply line.

---

## 4.3

## POSITIONING

---

When installing the machine, possible dangers that may arise from uncontrolled moving of the machine have to be kept in consideration. In these situations, it is necessary to block the machine to the floor.

### SPACE NECESSARY FOR USE AND MAINTENANCE

For check up and maintenance of the generator it is necessary to have the possibility to remove the front and the top panel.

Pay particular attention to air outlets for generator cooling; these outlets must be free and easily accessible for maintenance (filter cleaning etc.).

### ENVIRONMENTAL CONDITIONS ALLOWED

#### *Environmental working conditions*

The electronic generator can work at temperatures included between +5°C and +40°C, humidity levels of 15%-85% without condensation.

## 4.4

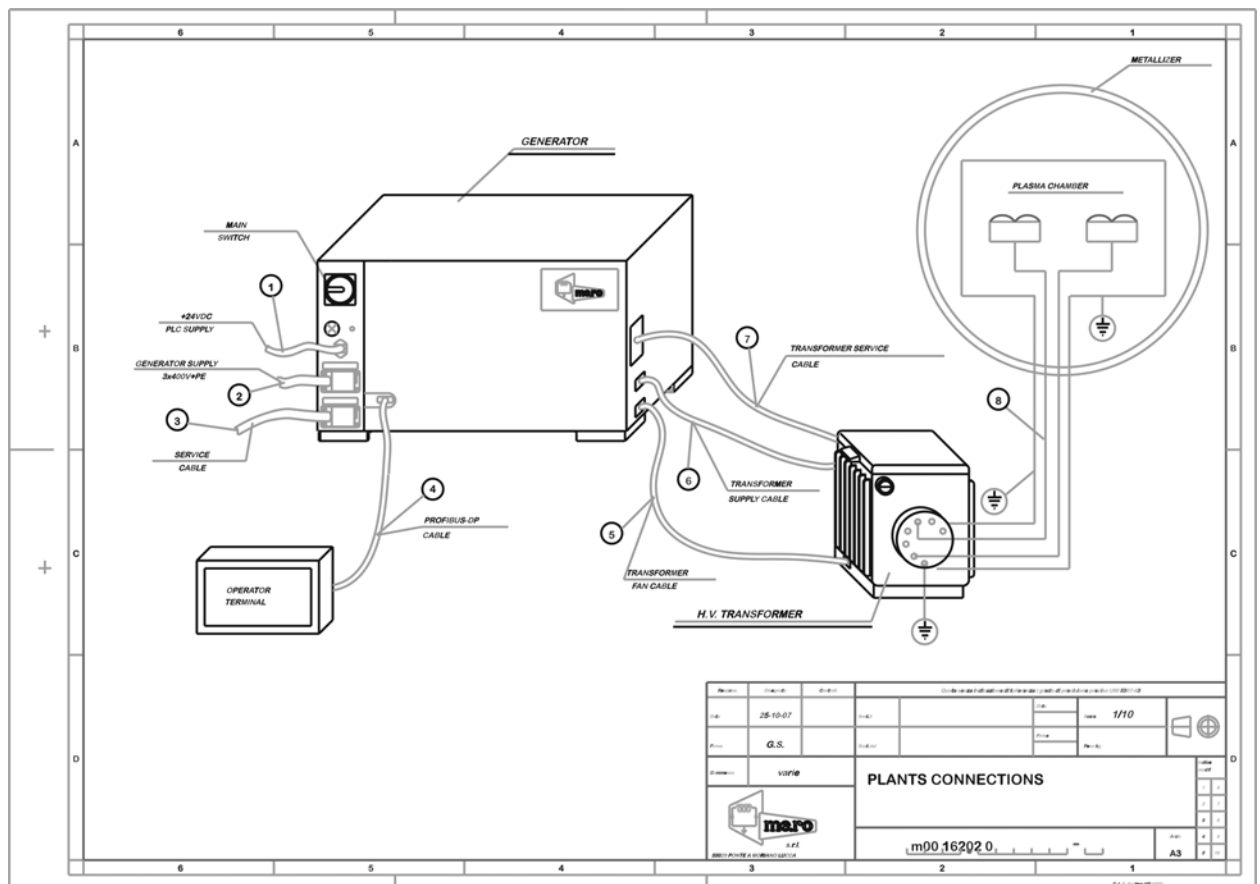
## CONNECTIONS



### ***Danger: high voltage!***

*The generator's electrical connections must be performed by qualified personnel, trained to work with electronic devices.*

These persons must follow local safety rules for the installation of the unit.





## POWER LINE

### *Supply line size*

Protections on the general supply panel (switch plus fuses) must be of the appropriate size, according to the supply voltage and the max. absorbed current by the generator; see INPUT electrical data label located on the back of the generator's cabinet.

## Specifications

### **ELECTRONIC GENERATOR "Solid State" MR\VP 10K-CH**

SUPPLY VOLTAGE: **3x380-460 V\* – 50-60 Hz**

**\*The voltage of the TR1 services transformer inside the generator must be the same as the supply voltage**

MAX. ABSORBED POWER: 11,5 kVA

MAX. OUTPUT POWER: 10 kW

WORK FREQUENCY: 40 - 50 kHz

AIR COOLING

DIMENSIONS WxHxD = 656 x 386 x 386

WEIGHT Kg. 39

### **TRANSFORMER MR\T-VP 10K-CH**

DIMENSION WxHxD = 396 x 403 x 320

WEIGHT Kg. 56

## 4.5

## INSTRUCTIONS FOR SETTING UP THE GENERATOR

Type MRVP 10K-CH (see connections drawing)

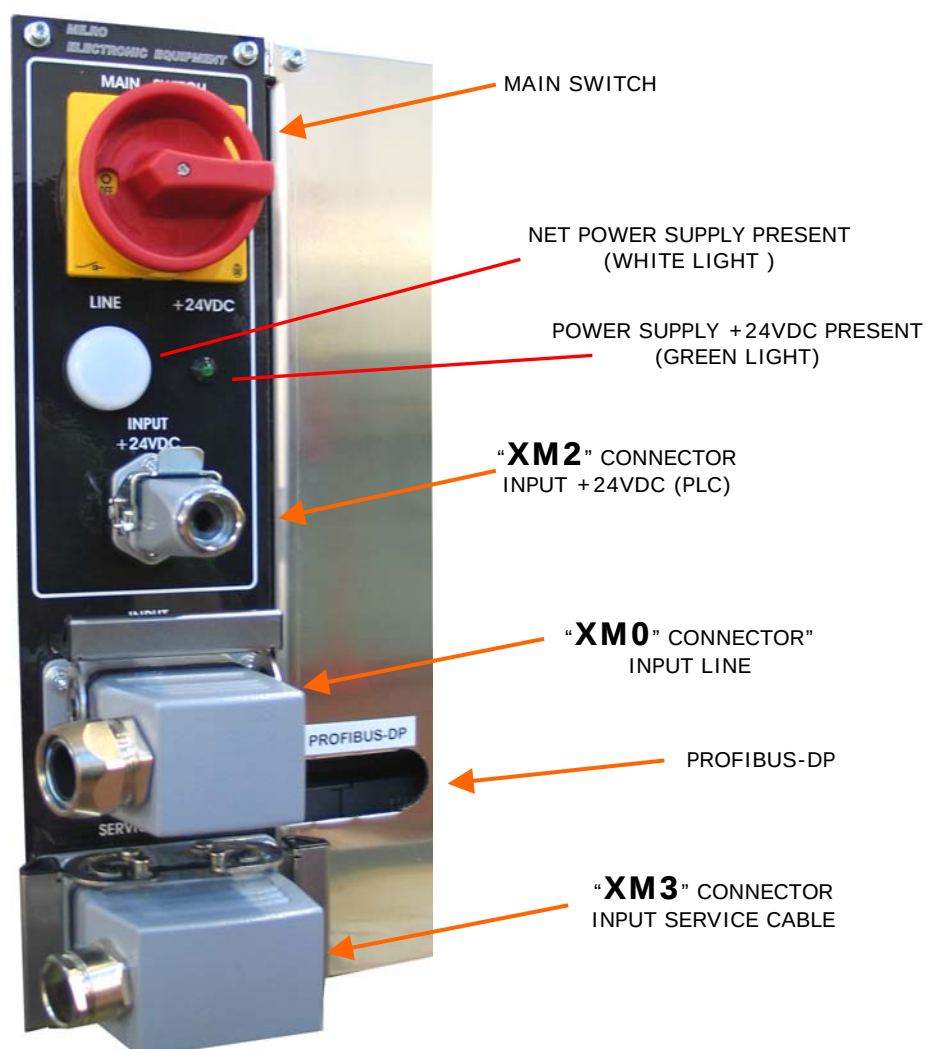


Fig.1 "Generator supply line"

### GENERATOR POWER SUPPLY

Following the electrical diagram, connect the connector **XF0** to the connector **XM0** fitted on the front panel of the chassis as in fig.1 (**INPUT LINE**)

**N.B.** The generator supply line must be protected by a differential switch to interrupt supply in case of loss of isolation.

The cable must be of appropriate size, keeping into account the max absorbed current (see INPUT data label located on the generator's cabinet).

### PLC S7-200 SIEMENS POWER SUPPLY

Connect the connector **XF2** to the connector **XM2** fitted on the front panel of the chassis as in fig.1 (**SUPPLY +24VDC**) (for PLC).

## SERVICES AND SAFETYS

Connect the connector XM3 to the connector **XF3** fitted on the front panel of the chassis as in fig.1 (**INPUT SERVICE CABLE**)

## OUTPUT TRANSFORMER SIGNALS

Connect the cable "SIGNAL CABLE" between connector **XM4** and the output transformer

## POWER SUPPLY OF THE OUTPUT TRANSFORMER FAN

Connect the cable "SUPPLY TRANSFORMER FAN CABLE" between connector **XF5** and the output transformer

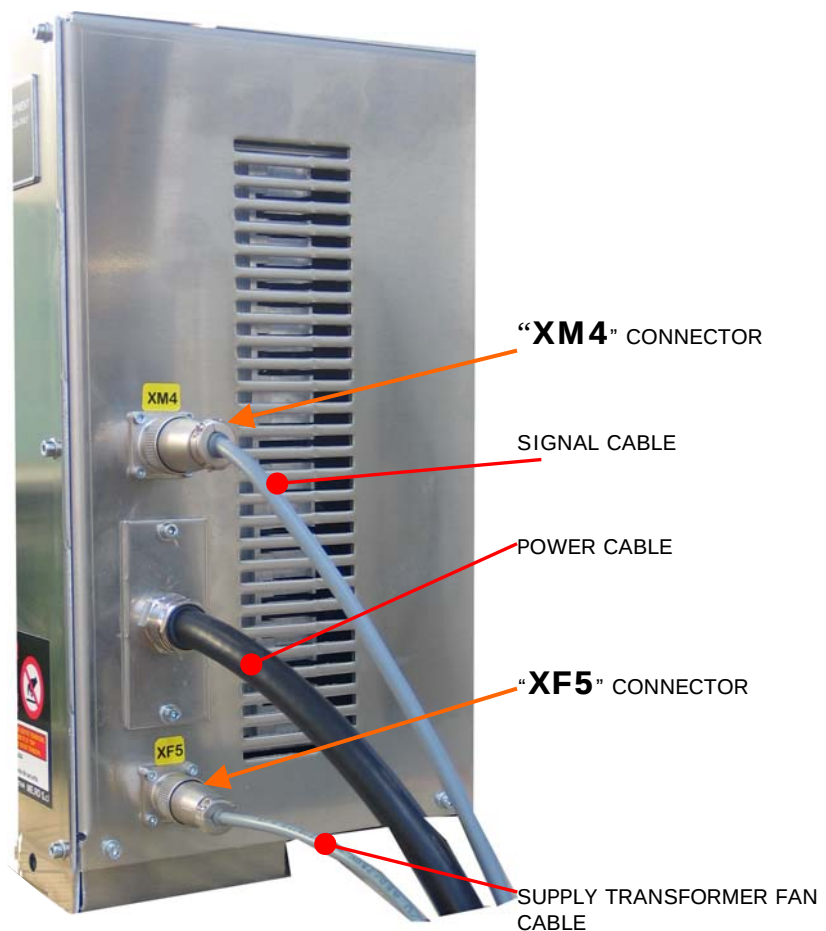


Fig.2 "Generator output lines"

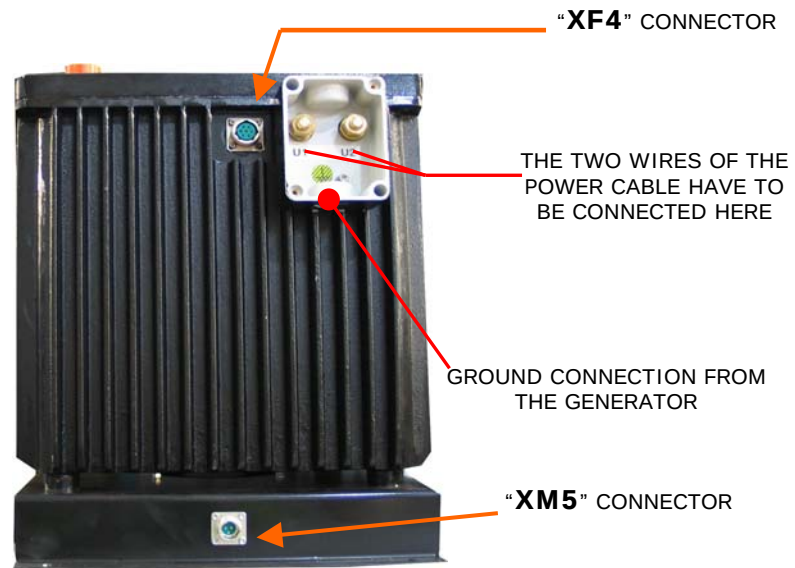


Fig.3 "Outlet Transformer"

**THE TWO WIRES OF THE POWER CABLE COMING FROM THE GENERATOR HAVE TO BE CONNECTED TO THE JUNCTION BOX OF THE OUTPUT TRANSFORMER.**



Fig.3 "Connections Transformer"

**THE GROUND WIRE HAS TO BE CONNECTED TO THE APPPOSITE SCREW INSIDE THE JUNCTION BOX OF THE OUTPUT TRANSFORMER.**



**DO NOT TURN ON THE GENERATOR BEFORE HAVING CONNECTED THE POWER SUPPLY CABLE TO THE OUTPUT TRANSFORMER .**

## 4.6

## ADAPTATION OF THE GENERATOR TO THE LOAD

Secondary taps of the  
output transformer



The data label on the output transformer indicates the taps and the relevant turn number. The turn number of each tap is also indicated on the white label put on the white insulator.

- Before turning the generator ON, **check that the vacuum conditions and the gas issued in the plasma chamber are those of standard operation.**
- Select the **manual** operation mode.
- Bring the setpoint for power adjustment to zero and turn the generator ON.
- Read the output values **PW** (VD2080), **IDC** (VD2066), **VDC** (VD2070) **PROFIBUS-DP** on the supervisor. Consider that the max. output power of the generator with the setpoint for power adjustment equal to **100 is 10kW**, and the values **IDC and VDC are respectively 25A. 400V.**
- Gradually increment the power adjustment setpoint up to 100 and check the generator's output power. If the generator is not correctly adapted to the load, it will not be able to issue the max power and in this case, two situations may arise:

1° The value of the VDC. reaches 400V. but the IDC current remains below 25A.

2° The value of the IDC reaches 25A. But the VDC voltage remains below 400V.

In the first case, it is necessary to utilize a higher numbers of turns on the output transformer's secondary.

In this way, the VDC value decreases while the IDC value increases.

In the second case, it is necessary to use a lower tap on the output transformer's secondary.

If necessary, in order to have a fine adjustment, it is possible to modify the variable impedance which is serial connected to the primary coil of the output transformer ( see electric diagram of the generator). The impedance is fitted inside the output transformer box and of course this adjustment has to be made by qualified ME.RO personnel.

In this way, the VDC value increases while the IDC value decreases.

If max output power is not required, the generator can correctly work also with IDC VDC values different from the max limit of both.

The max value of the individual sizes is in any case limited for the VDC to 400V and for the IDC to 25A.

Summarizing, optimal working conditions for the generator with max output power are:

**PW = 10 kW**

**VDC = 400 V.**

**IDC = 25 A.**

These values are obtained by using the most appropriate tap on the output transformer's secondary so as to obtain optimal match between generator and load.

If the plasma chamber is not able to work with the generator's max output power, or if the metallization process does not require max power, the

---

generator can work without any problems (as already mentioned above)  
even if optimal matching conditions are not reached

---

## 5 OPERATION, CONTROL

---

*Who is this chapter directed at?*

### Profile of this chapter:

This chapter is directed to persons who utilize the electronic generator.

*Contents of the chapter*

This chapter contains a description of the generator's operational modes and states.

---

### 5.1 SETTING UP

---

The generator has been tested by *Me.Ro. srl* at its premises.

It can be set up by following what explained in paragraphs 3.2 4.4 4.5

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### 5.2 STRUCTURE OF THE OPERATOR TERMINAL

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In this specific application, the operator terminal is replaced by specific pages of the supervision system. Therefore, to set the variables and the commands, the operator must refer to the specific instructions of the supervisor.

---

### 5.3 TURNING THE GENERATOR ON/OFF

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#### CHECKING SAFETY DEVICES



*Before turning the generator on, check the following points:*

- All safety elements must be active
- All remote interlocks must be closed

#### PREPARATION FOR OPERATION



*Supplying the generator.*

*Checking the following points:*

- Check the alarm signal menu on the operator terminal.
- Remove any alarms present

NB. The generator may be turned on only after having eliminated all alarms.



## 5.4 START OF OPERATION

The generator is turned ON by the supervision program.

## 5.5 TURNING THE GENERATOR OFF

The generator is turned OFF by the supervision program.

## 5.6 TURNING OFF IN CASE OF EMERGENCY



*Press the emergency button.*



### **Danger: high voltage!**

After having pressed the emergency button, the generator may have residual voltages (see chapter 2: "Safety")

## 5.7 SETTING PARAMETERS



Before turning the system on, you must set some parameters on the Operator Terminal (supervision) using the following page.

### **Remote Control.**

Bit Profibus – DP enabling \*\*\*\*

|                |                    |         |
|----------------|--------------------|---------|
| REMOTE CONTROL | 1 = REMOTE ENABLED | V2000.2 |
|----------------|--------------------|---------|

### **Operation mode.**

Bit Profibus – DP selection

|                    |                                 |         |
|--------------------|---------------------------------|---------|
| SEL MAN.           | 1 = MANUAL MODE SELECTED        | V2000.3 |
| SEL AUT.           | 1 = AUTOMATIC MODE "W" SELECTED | V2000.4 |
| SEL AUT SPEC POWER | 1 = AUT SPEC POWER SELECTED     | V2000.5 |

**Low Speed.** Setting the minimum speed below which the generator turns to OFF. Word Profibus – DP setting

|           |                 |        |
|-----------|-----------------|--------|
| LOW SPEED | LOW SPEED m/min | VW2002 |
|-----------|-----------------|--------|

**Power max. deviation.** Setting the max output power deviation % admissible before enabling the signalling “Power too High” or “Power too Low” (see message and signalling alarms paragraph 5.9). Word Profibus – DP setting

|                     |                                     |        |
|---------------------|-------------------------------------|--------|
| POWER MAX DEVIATION | MAX POWER DEVIATION<br>ADMISSIBLE % | VW2004 |
|---------------------|-------------------------------------|--------|

**Plasma chamber length.** Setting plasma chamber length. (This datum, together with speed, is used to calculate production in m<sup>2</sup>/min.) Word Profibus – DP setting

|                       |                              |        |
|-----------------------|------------------------------|--------|
| PLASMA CHAMBER LENGTH | PLASMA CHAMBER LENGTH<br>cm. | VW2006 |
|-----------------------|------------------------------|--------|

**Power control.** Setting the Setpoint to adjust output power. Word Profibus – DP setting

|                   |                                                                   |        |
|-------------------|-------------------------------------------------------------------|--------|
| SP MAN            | SETPOINT MANUALE 0 100 %                                          | VW2008 |
| SP AUT            | SETPOINT AUTOMATICO<br>1000 10000 W                               | VD2010 |
| SP AUT SPEC POWER | SETPOINT AUTOMATIC<br>SPECIFIC POWER Wmin/m <sup>2</sup> 5-<br>50 | VW2014 |

**Speed.** Speed of work

|       |                        |        |
|-------|------------------------|--------|
| SPEED | SPEED OF WORK (number) | VW2016 |
|-------|------------------------|--------|

**Number of arcs.** Setting the number of admissible arcs in the opening time of the count window (See Arc Management paragraph 3.1). Word Profibus – DP setting

|                |                       |        |
|----------------|-----------------------|--------|
| NUMBER OF ARCS | SELECT NUMBER OF ARCS | VW2018 |
|----------------|-----------------------|--------|

**Time for arcs.** Setting time in seconds for the arc count window. (See Arc Management paragraph 3.1). Word Profibus – DP setting

|               |                          |        |
|---------------|--------------------------|--------|
| TIME FOR ARCS | TIME FOR ARCS IN SECONDS | VW2020 |
|---------------|--------------------------|--------|

**Tap used.** Setting the secondary turns used on the output transformer (this datum is used to calibrate the reading of the Vout rms). Word Profibus – DP setting

|          |                        |        |
|----------|------------------------|--------|
| TAP USED | OUT.TRANSF. TURNS USED | VW2022 |
|----------|------------------------|--------|

## 5.8 OPERATION MODE

Power control on the generator can be performed in three different ways.

**Manual mode.**

**Automatic mode W**

**Automatic specific power mode.**

Selection is made by acting on bits 3 - 4 - 5 of V2000 (Profibus-DP).

|                          |                              |         |
|--------------------------|------------------------------|---------|
| MANUAL MODE              | 1 = SEL. MANUAL              | V2000.3 |
| AUTOMATIC MODE W         | 1 = SEL. AUT.                | V2000.4 |
| AUTOMATIC SPECIFIC POWER | 1 = SEL. AUT.<br>SPEC. POWER | V2000.5 |

### Manual mode

In the manual mode, the generator's output power can be regulated from minimum to maximum through the value "**SP MAN 0 100 %**" inserted in Word **VW2008** (Profibus-DP)

### Automatic mode W

In the automatic mode of operation, output power is regulated by setting the setpoint with the desired output power value. "**SP AUT. 1000 10000 W**" inserted in word **VD2010** (Profibus – DP)

By using this type of operation, any load deviations due for example, to a variation in the vacuum degree inside the chamber, or to a change in the quantity of gas issued with consequent change in the output power, are automatically compensated for.

### Automatic specific power mode

In this operation mode, the PLC's CPU calculates the specific power issued by the generator "**Wmin/m<sup>2</sup>**", compares it with the "**SP AUT SPEC POWER 5 - 50 Wmin/m<sup>2</sup>**" inserted in word **VW2014** (Profibus – DP), and, through an adjustment algorithm, maintains it constant (equal to the set value).

Using this operation mode, once the specific power to be used has been determined, it is maintained constant even with changes in working speed and in the load resistance of the plasma chamber.

Any conditions that are beyond the control of the adjustment algorithm are signaled through messages "**POWER TOO HIGH or POWER TOO LOW**"

If we insert a specific power value that is too high, we will see the signal **POWER TOO LOW** because in order to maintain the set specific power, the generator should issue more than the maximum admissible power.

On the contrary, if we insert a specific power value that is too low, we will see the signal **POWER TOO HIGH** because the generator's output power cannot in any case go below a minimum value.

**We therefore advise to set the SP AUT SPEC POWER to a minimum value that is not too low in order to avoid (at low working speeds) adjustment loss.**

**Specific Power in  $Wmin/m^2$**  is obtained through the formula:

**Spec. Power =  $P_{out}/Production$**

where

**$P_{out} = I_{Dc} \times V_{Dc}$**

**Production = Plasma chamber length x Speed.**

The value "**Spec. Power =  $Wmin/m^2$** " is present in Word **VD2084**  
(Profibus - DP)

The value " **$I_{Dc} = A.$** " is present in Word **VD2066**  
(Profibus - DP) The value " **$V_{Dc} = V.$** "  
is present in Word **VD2070** (Profibus - DP)

The value "**Speed. = m/min.**" is present in Word **VD2094**  
(Profibus - DP)

The value "**Lungh.Plasma Chamber = cm.**" is present in word **VW2006**  
(Profibus-DP)

## 5.9

## ALARM AND SIGNAL MESSAGES

### *Alarm messages*

Alarm messages are the following:

- **LOW SPEED:** Production speed inferior to the speed set in the Set Parameters page on the operator terminal.
- **EMERGENCY:** Emergency button pressed.
- **OVERLOAD:** Intervention of peak current limitation due to a generator overload or to having exceeded the number of arcs with respect to the number set on the Set Parameters page of the operator terminal.
- **Vcc FAULT:** Lack of continuous supply voltage at the Chopper stage.
- **THERMAL OVERLOAD:** Intervention of the generator's thermal protections.
- **COOLING WATER PLASMA CHAMBER:** Lack of cooling liquid in the plasma chamber.
- **ARC LIMIT REACHED:** the arc number has exceeded the value which has been set in word 2018 .
- **METALLIZER OPEN:** Metallizer safety interlock is open

### SIGNAL MESSAGES

- **LOW SPEED:** Production speed inferior to the speed set on the Set Parameters page of the operator panel.
- **READY TO START:** No alarm in the generator is ready for ON.
- **PRESENT ALARM:** An alarm that is keeping the generator OFF is present.
- **GENERATOR ON:** Generator in operation.
- **POWER TOO HIGH:** Increase in the output power beyond the preset limit % (Admissible Power Dev.) in the Set Parameters page on the Operator Panel.
- **POWER TOO LOW:** Decrease in the output power beyond the preset limit % (Admissible Power Dev.) in the Set Parameters page on the Operator Panel.

The above messages are represented by the absolute value assumed by the **Word Alarm VW2084** and by the **Word Signalling VW2086**.

See paragraph 3.2 Profibus – DP interface table

The work setpoint deviation causes the intervention of the “Power too High” or “Power too Low” alarm signalling, without blocking the generator. The signalling must be restored by supplying an impulse on the V2000.1 (see Profibus DP table).

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## 6 MAINTENANCE

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*Who is this chapter directed at?*

**Profile of this chapter:**

This chapter is directed to persons perform maintenance work on the electronic generator.

*Contents of the chapter*

This chapter contains advice and notices regarding the generator's maintenance.

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### 6.1 GENERAL NOTES ON MAINTENANCE

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In this chapter you will find maintenance advice that describes the work necessary in order to maintain the generator's proper operation through time.

Extraordinary maintenance should be performed by *Me.Ro. srl*'s technical personnel.

**TECHNICAL SUPPORT**

*Me.Ro. srl*'s technical personnel can be contacted via phone at the company's office numbers.

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### 6.2 SAFE MAINTENANCE

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*To protect one's own safety, we advise that you follow the safety rules and measures described here following each time the generator requires maintenance.*

**DANGEROUS AREAS**



***Danger: high voltage!***

*The generator converts low frequencies in DC energy. Particulars under high voltage may cause electrical shock.*



Ensure that you are well acquainted with the generator's dangerous areas. Carefully read chapter 2.3 "Dangerous areas: Maintenance and repair"

#### REMOVAL OF RESIDUAL VOLTAGE

##### ***Danger: high voltage!***



After having turned the generator off, some parts may present residual voltage (see chapter 2: "Safety")



Disconnect the generator and follow the safety norms concerning high electrical voltage systems.

### 6.3

#### MAINTENANCE INTERVALS

##### *Regular maintenance*

Regularly inspect the RF generator so as to be able to operate it for long periods of time.

Maintenance work should be performed keeping in mind the working time and the environmental conditions under which the generator operates.

### 6.4

#### DISMANTLING AND PACKING

##### UNIT ISOLATION

1. Turn the generator off
2. Interrupt the generator's supply line
3. Remove all external electrical connections

Remove all connections for water flow and completely empty the cooling circuit. Remove all the water from the circuit also with the aid of compressed air.



##### ***Danger!***

*Make the cooling water flow off in the opposite direction with respect to the normal working flow*

##### PACKING



Pack the generator according to its dimensions.

Place it on a pallet, cover it with plastic film and wrap it with belts.



##### TRANSPORT

See chapter 5.1 "Transport"

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## DISPOSAL

The generator is composed mainly by.

Metals:

- a. Aluminium
- b. Copper
- c. Steel
- d. Brass
- e. Galvanized iron sheets
- f. Steel sheets

Plastics:

- g. Teflon , PA and others

Other materials:

- h. Electronic cards
- i. Ceramic



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Cyclostyled sheet owner in gennaio '08

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